

Building Multi-Step Arguments from Givens

Answer Key

High-school geometry

Quick Answers

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|---|---|
| 1. — | 6. (b) the overlapping distance = 15.2 cm, — |
| 2. — | 7. (b) $x = 15$, — |
| 3. (a) $x = 4$, (b) the whole length = 30 cm | 8. (a) $x = 4$, (b) the whole length = 28 cm |
| 4. (a) $x = 20$, (b) the vertical angle = 75 degrees | 9. (b) $x = 12$, — |
| 5. — | 10. (b) $x = 15$, — |
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What is verified

3 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 5, 6, 7, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: High-school geometry

HSG-CO.A – problems 2, 3, 8

HSG-CO.C – problems 1, 4, 5, 6, 7, 9, 10

Difficulty 1–5 of 5 across 17 tagged checks.

1. Order the four statements about $\angle FGH$ and $\angle HGK$ and give the reasons.

Order and reasons:

1. $\overrightarrow{GF}$ and $\overrightarrow{GK}$ are opposite rays, and $\overrightarrow{GH}$ is a third ray from G — *Given*.
2. $\angle FGH$ and $\angle HGK$ form a linear pair — *Definition of a linear pair* (adjacent angles whose non-common sides are opposite rays).
3. $m\angle FGH + m\angle HGK = 180$ — *Linear Pair Postulate*.
4. $m\angle FGH = 180 - m\angle HGK$ — *Subtraction property of equality*.

Grading: the order is forced by the reasons: nothing before line 1 is available, and line 4 can only come from line 3. Full credit needs both the order and the four reasons.

2. Order the five statements proving a midpoint halves a segment.

Order and reasons:

1. M is the midpoint of $\overline{AB}$ — *Given*.
2. $\overline{AM} \cong \overline{MB}$ — *Definition of midpoint*.
3. $AM = MB$ — *Definition of congruent segments*.

4. $AM + MB = AB$ — *Segment Addition Postulate*.

5. $2AM = AB$, so $AM = \frac{1}{2}AB$ — *Substitution, then the division property of equality*.

Grading: lines 2 and 3 are two different reasons and must be kept apart — the first gives congruence, the second turns congruence into an equation of lengths.

3. M midpoint of $\overline{AB}$, $AM = 4x - 1$, $MB = 2x + 7$ cm. (a) x . (b) AB .

(a) Midpoint gives $\overline{AM} \cong \overline{MB}$, so the lengths are equal:

$$4x - 1 = 2x + 7$$

$$2x = 8$$

So $x = 4$.

$$\boxed{x = 4}$$

(b) $AM = 4(4) - 1 = 15$ and $MB = 2(4) + 7 = 15$, so by the Segment Addition Postulate $AB = 2AM = 2(15)$.

$$\boxed{AB = 30 \text{ cm}}$$

4. Intersecting lines at V : $m\angle AVB = 3x + 15$, $m\angle BVC = 5x + 5$, linear pair. (a) x . (b) $m\angle CVD$.

(a) The Linear Pair Postulate gives a sum of 180:

$$(3x + 15) + (5x + 5) = 180$$

$$8x + 20 = 180$$

$$8x = 160$$

So $x = 20$. Check: $75 + 105 = 180$.

$$\boxed{x = 20}$$

(b) $\angle CVD$ is vertical to $\angle AVB$, so the two are congruent and have equal measures: $m\angle CVD = m\angle AVB = 3(20) + 15$.

$$\boxed{m\angle CVD = 75^\circ}$$

5. Order the statements proving complements of the same angle are congruent.

Order and reasons:

1. $\angle P$ and $\angle Q$ are complementary; $\angle R$ and $\angle Q$ are complementary — *Given*.

2. $m\angle P + m\angle Q = 90$ and $m\angle R + m\angle Q = 90$ — *Definition of complementary angles*.

3. $m\angle P + m\angle Q = m\angle R + m\angle Q$ — *Substitution* (both sums equal 90).

4. $m\angle P = m\angle R$, so $\angle P \cong \angle R$ — *Subtraction property of equality*, then the *definition of congruent angles*.

Grading: the substitution step is the one students drop; without it there is no link between the two separate sums.

6. A, B, C, D in order with $AB = CD$. (a) Argue $AC = BD$. (b) With $AB = CD = 4.2$ cm and $BC = 11$ cm, find AC .

(a) **Model argument:**

1. $AB = CD$ — *Given*.

2. $AB + BC = BC + CD$ — *Addition property of equality* (the shared stretch BC added to both sides).

3. $AB + BC = AC$ and $BC + CD = BD$ — *Segment Addition Postulate*; therefore $AC = BD$ by *substitution*.

Grading: the argument must add the shared BC with a named property and rename both sums with the postulate. “The pieces are equal so the wholes are equal” with no reasons is partial credit.

(b) $AC = AB + BC = 4.2 + 11.$

$AC = 15.2 \text{ cm}$

7. $\overrightarrow{QS}$ bisects $\angle PQR$; $\angle PQR, \angle RQT$ a linear pair with $m\angle RQT = 110$; $m\angle PQS = 2x + 5$.
 (a) The chain. (b) x .

(a) **The chain:**

1. $\angle PQR$ and $\angle RQT$ form a linear pair — *Given*; so $m\angle PQR + 110 = 180$ — *Linear Pair Postulate*.

2. $\overrightarrow{QS}$ bisects $\angle PQR$ — *Given*; so $m\angle PQR = 2m\angle PQS$ — *Definition of an angle bisector* (with the Angle Addition Postulate).

3. Substituting $m\angle PQS = 2x + 5$ into step 2 and that result into step 1 gives $2(2x + 5) + 110 = 180$ — *Substitution*.

(b)

$$2(2x + 5) + 110 = 180$$

$$4x + 10 + 110 = 180$$

$$4x = 60$$

So $x = 15$. Check: $m\angle PQS = 35$, $m\angle PQR = 70$, and $70 + 110 = 180$.

$x = 15$

8. N midpoint of $\overline{RT}$, $RN = 3x + 2$, $NT = x + 10$ cm. (a) x . (b) RT .
 (a)

$$3x + 2 = x + 10$$

$$2x = 8$$

So $x = 4$.

(b) $RN = 3(4) + 2 = 14$ and $NT = 4 + 10 = 14$, so $RT = 2RN = 2(14)$.

$x = 4$

$RT = 28 \text{ cm}$

9. $\angle J, \angle K$ complementary; $\angle L, \angle K$ complementary; $m\angle J = 4x - 6$, $m\angle L = 2x + 18$.
 (a) Argue $\angle J \cong \angle L$. (b) Find x .

(a) **Model argument:**

1. $\angle J$ and $\angle K$ are complementary; $\angle L$ and $\angle K$ are complementary — *Given*.

2. $m\angle J + m\angle K = 90$ and $m\angle L + m\angle K = 90$ — *Definition of complementary angles*.

3. $m\angle J + m\angle K = m\angle L + m\angle K$ — *Substitution*.

4. $m\angle J = m\angle L$, so $\angle J \cong \angle L$ — *Subtraction property of equality* and the *definition of congruent angles*.

(b) The result of (a) lets you set the two expressions equal:

$$4x - 6 = 2x + 18$$

$$2x = 24$$

So $x = 12$. Check: $m\angle J = 42$ and $m\angle L = 42$.

$x = 12$

10. $\overrightarrow{QS}$ bisects $\angle PQR$; $\angle PQR, \angle RQT$ a linear pair; $m\angle PQS = 2x + 15$, $m\angle RQT = 6x$.

(a) The whole argument. (b) x .

(a) Model argument:

1. $\overrightarrow{QS}$ bisects $\angle PQR$ — *Given*.

2. $m\angle PQR = 2m\angle PQS = 2(2x + 15)$ — *Definition of an angle bisector* with the Angle Addition Postulate, then *substitution*.

3. $\angle PQR$ and $\angle RQT$ form a linear pair, so $m\angle PQR + m\angle RQT = 180$ — *Given* and the *Linear Pair Postulate*.

4. $2(2x + 15) + 6x = 180$ — *Substitution* of steps 2 and the given $m\angle RQT = 6x$ into step 3.

(b)

$$2(2x + 15) + 6x = 180$$

$$4x + 30 + 6x = 180$$

$$10x = 150$$

So $x = 15$. Check: $m\angle PQS = 45$, $m\angle PQR = 90$, $m\angle RQT = 90$, and $90 + 90 = 180$.

$x = 15$

Grading (a): both givens used, the bisector definition doubling the marked half, the linear pair postulate supplying the sum, and a substitution that lands on one equation in x .